

INSECT BITES AND STINGS WITH URTICARIA (HIVES)

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Pharm*Achieve*

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DISCLAIMER

- While progressing through this learning and using any of the resources provided, please keep in mind:
 - ✓ the scope of practice for pharmacists within your province or territory
 - ✓ the policies and procedures of your designated place of work
 - ✓ your scope of practice may be more limited or extensive than presented in this eLearning module
- The treatment examples provided are intended to illustrate possible indicated options for managing the specified common/minor ailment or condition. Pharmacists should:
 - ✓ follow provincial/territorial regulations regarding treatment options they are authorized to prescribe
 - ✓ exercise professional judgment at all times to prescribe the most appropriate option for each patient
 - ✓ refer to the latest clinical guidelines and product monographs

LEARNING OBJECTIVES

INSECT BITES AND STINGS WITH URTICARIA (HIVES)



Describe the clinical presentation and pathophysiology of insect bites and stings with urticaria



Identify possible risk factors for insect bites and stings with urticaria



Recognize potential differential diagnoses and red flags requiring a referral for insect bites and stings with urticaria



Describe the non-pharmacological and pharmacological treatment options for insect bites and stings with urticaria



Account for considerations associated with treating insect bites and stings with urticaria in special populations



Understand monitoring parameters for insect bites and stings with urticaria



LEGEND

DEET: N,N-diethyl-meta-toluamide

N/V: Nausea and/or Vomiting

PMD: Para-Menthane-3,8-Diol

S/E: Side Effects

WNV: West Nile Virus

A top-down view of a doctor's desk. A person in a white lab coat is typing on a silver laptop. To the right is a blue stethoscope on a white clipboard. Below the laptop is a smartphone. To the left is a silver keyboard. The desk is light-colored wood. There are some geometric overlays on the image.

CLINICAL PRESENTATION & ASSESSMENT



CLINICAL PRESENTATION

- Localized skin reaction
 - Pruritus
 - Erythema
 - Edema
- Mild allergic reaction
 - Urticaria: 1 to 5 cm in size, pruritic, raised, red bumps
 - May have minimal bleeding if scratched or manipulated
- Severe allergic reactions involving anaphylaxis require immediate emergency medical attention 



ANAPHYLAXIS

- Life-threatening allergic reaction with rapid progression | 
 - Difficulty breathing, shortness of breath, wheezing
 - Severe swelling, urticaria, flushing
 - Hypotension, N/V or diarrhea, dizziness or fainting
- Can occur within minutes or up to hours after exposure
- Can be biphasic, with a second-phase reaction occurring as late as 72 hours after the initial reaction is resolved
- Any signs of anaphylaxis should be treated immediately by administration of epinephrine, and the patient should be promptly referred for emergency care



ETIOLOGY OF HIVES



Insect bites or stings



Acute viral infections



Allergic reactions to food or medications



CAUSATIVE INSECTS

Bites

- Common insects that cause bites include mosquitoes, flies, fleas, and ticks
- Other examples: spiders, bed bugs, ants, lice

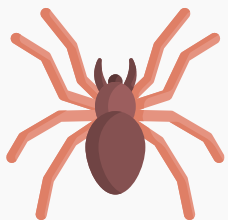
Stings

- Common insects that cause stings include ants, bees, hornets, and wasps
- Bee stings are the most common cause of anaphylaxis resulting from a sting

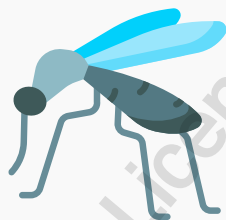


IMAGES OF INSECT BITES AND STINGS

Spider



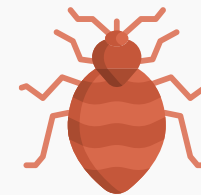
Mosquito



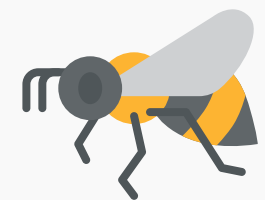
Tick



Bed bug



Bee





IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Patient History:

The patient has just returned from a trip to Iceland where they visited waterfalls and stayed at a local hotel.





IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Answer: Bed bugs

SYMPTOMS

- Mild to severe pruritus is the primary symptom
- Mild macular spots to larger wheals
- Spots may last 7 to 14 days
- Minor bleeding may occur
- May lead to insomnia and/or anxiety





IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Patient History:

The patient is a photographer that commonly visits local flower fields to take pictures.





IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Answer:
Bee/wasp/hornet



SYMPTOMS

- Initial symptoms of sharp or burning pain and skin swelling and redness
- Raised bump with red in the centre
- Erythema surrounding the area
- May have extensive swelling not considered anaphylaxis



IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Patient History:

The patient recently had a picnic with her children in a local park.



Image courtesy of Bart Everson. [License](#).
Image was cropped



IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Answer:
Fire/red ant



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SYMPTOMS

- Initial symptoms of burning, redness, and possible localized swelling
- Pustules may form within a day and resolve within 2 weeks
- Stings may leave a scar when healed



IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Patient History:

The patient just returned from a camping trip at a campsite located beside a lake.





IMAGES OF INSECT BITES AND STINGS

What insect caused these reactions?

Answer:
Mosquito



SYMPTOMS

- Initial symptoms of pruritus, redness, and localized swelling
- Bites will usually self-resolve in 24 to 48 hours
- Mosquitoes are known to be vectors for West Nile virus, yellow fever, dengue fever



PATHOPHYSIOLOGY

- Mast cells play an important role in urticaria
- Activation causes degranulation which releases:
 - Histamine
 - Leukotrienes
 - Prostaglandins
- Acute urticaria (hives lasting <6 weeks) may occur spontaneously or from an allergy-induced reaction



PATHOPHYSIOLOGY

- Venom from insects contains proteolytic enzymes and high molecular weight proteins, which act as allergens
 - Enzymes degrade tissue and cause a localized reaction, whereas high molecular weight proteins may cause a systemic reaction
- Pruritus results from a mediator-induced sensory nerve impulse, which turns into a motor impulse that tells us to scratch



WEST NILE VIRUS

- West Nile virus is one of the most common diseases transmitted by mosquitoes
- Symptoms may range from mild to severe
 - Fever, headache, stiff neck, N/V, muscle weakness
- 70% to 80% of those infected are asymptomatic
- Assessment
 - Based on symptoms, geographic location, time of year, and laboratory results
- Treatment
 - There is no specific treatment or vaccine for WNV
 - Analgesics, antipyretics for mild symptoms
- Long-term effects are not fully understood

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

What are common symptoms associated with insect bites? Select all that apply.

- ☐ a) Pruritus
- ☐ b) Erythema
- ☐ c) Edema
- ☐ d) Runny Nose

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

Is the following statement true or false?

Mast cells play an important role in urticaria, and release histamine, leukotrienes, and adrenaline to cause effects such as pruritus.

☐ True

☐ False



RISK FACTORS & PREVENTATIVE MEASURES



Outdoor environmental exposure



Close proximity to a hive or nest



Areas with tall grass, swamps, or bushy areas, especially at dawn or dusk



Use of scented products that may attract insects



Clothing that does not sufficiently cover exposed skin



DIFFERENTIAL DIAGNOSES

CONDITION	DESCRIPTION
Other allergic reactions (e.g. chemicals, food, medication)	Other identifiable allergen
Fungal skin infections	Lesions with raised, circular, and scaly edges
Impetigo	Nonbullous or bullous, blistering or crusting
Scabies	Severe itching that is often worse at night, burrows
Dermatitis	Skin dryness, lichenified plaques
Cellulitis	Swelling, pain, and inflammation at site



WHEN TO REFER

SITUATIONS WARRANTING PATIENT REFERRAL

- Any sign of anaphylaxis
- Pronounced inflammation of the tongue or mouth
- History of severe reactions to previous bites/stings
- Unresolved symptoms after >7 days of treatment
- Significant pain, bleeding, or swelling
- Rapidly spreading reaction
- Systemic symptoms (e.g. fever, N/V)
- Signs and symptoms of West Nile virus

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

Which of the following are risk factors for insect bites and urticaria? Select all that apply.

- ☐ a) Close proximity to areas with tall grass, swamps, or bushy areas at dawn or dusk
- ☐ b) Clothing that does not sufficiently cover exposed skin
- ☐ c) Proper use of insect repellents
- ☐ d) Use of scented products

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

JE presents with itchy skin on his torso. JE says these symptoms started on the weekend, after playing volleyball on a hot summer day. JE describes the area as a red rash with raised, circular, and scaly edges. What are these symptoms most consistent with? Select the correct answer option.

- ☐ a) Tick bites
- ☐ b) Bee stings
- ☐ c) Head lice
- ☐ d) Fungal skin infection

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

Which of the following patient cases warrants a referral? Select all that apply.

- ☐ **a)** A patient presenting with an itchy, localized skin reaction after being bitten by a mosquito
- ☐ **b)** A patient who was stung by a bee and presents with severe swelling and difficulty breathing
- ☐ **c)** A patient who was bitten by several ants and now presents with worsening symptoms despite 10 days of treatment
- ☐ **d)** A patient presenting with an itchy, localized skin reaction after being bitten by a spider



ASSESSMENT

- The World Allergy Organization does not recommend routine diagnostic measures in acute urticaria
- Cutaneous provocation tests may identify specific allergens
- A positive insect sting may be assessed if the stinger is visible

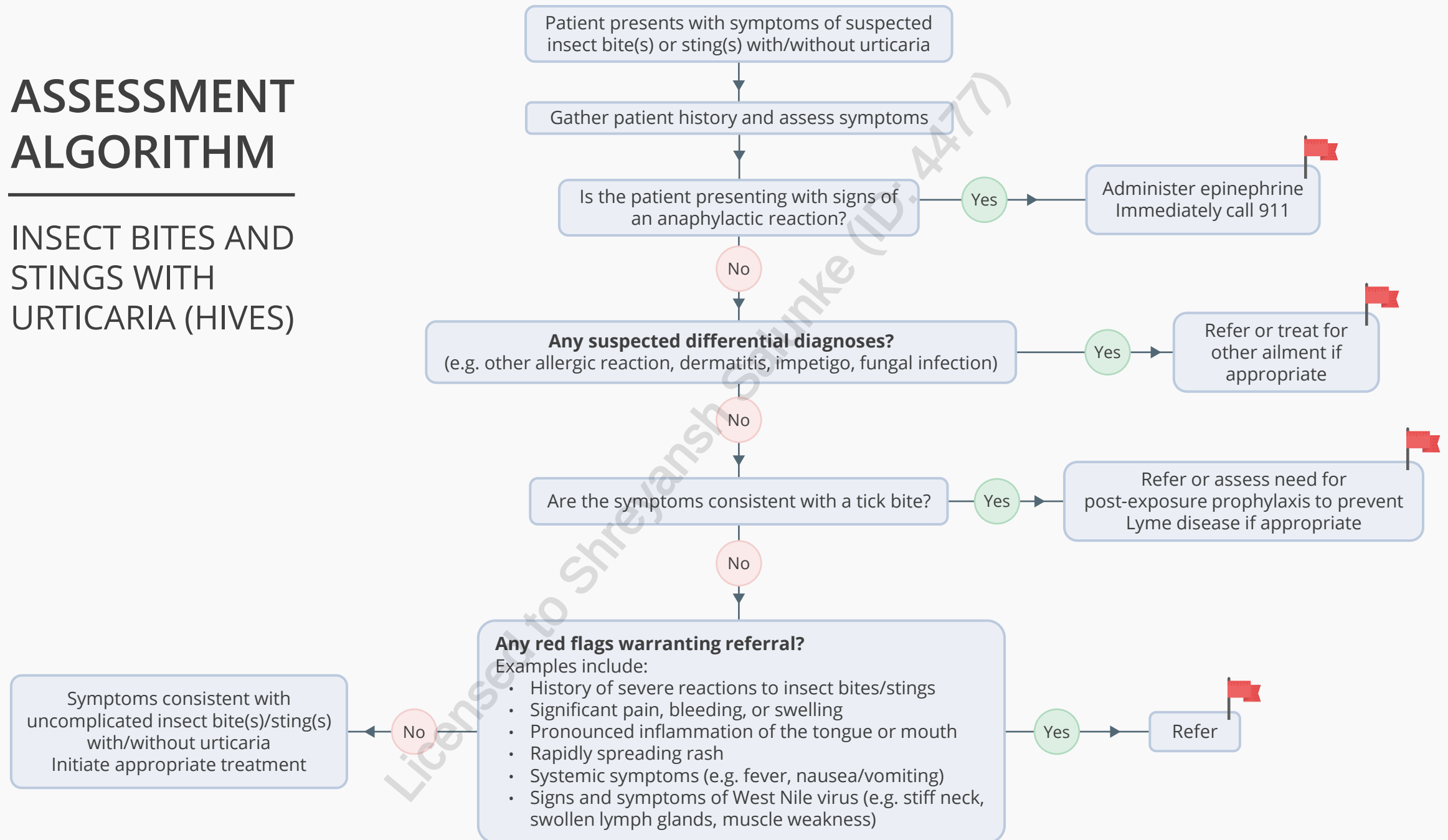


ASSESSMENT

- **Assess symptomatic history**
 - Pruritus, erythema, edema, mild allergic reaction?
- **Etiology: What kind of insect?**
- **What other symptoms are present?**
 - Anaphylaxis requires immediate administration of epinephrine and referral
- **Identify any red flags and complications**
 - History of severe reactions? Signs of West Nile virus?
- **Consider special populations**
 - Is the patient pregnant or breastfeeding?

ASSESSMENT ALGORITHM

INSECT BITES AND STINGS WITH URTICARIA (HIVES)

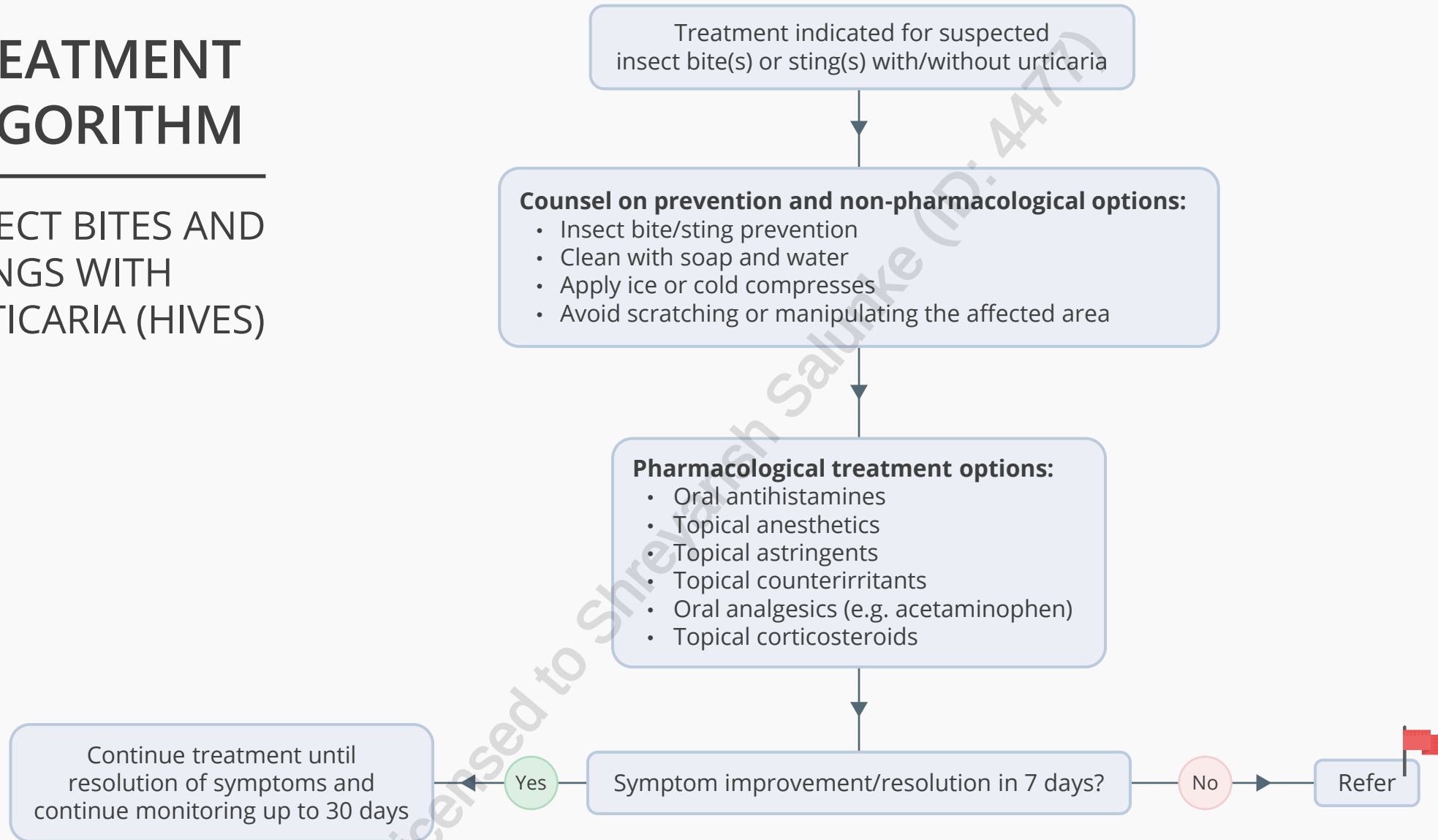




TREATMENT

TREATMENT ALGORITHM

INSECT BITES AND STINGS WITH URTICARIA (HIVES)





PREVENTION

- Avoid insect-infested areas, especially marshes, swamps, and tall bushes
- Reduce skin exposure and wear light-coloured, long-sleeved clothing
- Avoid outdoor activity at dawn or dusk
- Window screens, mesh netting, and bed nets
- Regular inspection of household items or items brought into the home
- Use DEET or icaridin as insect repellents (≥ 6 months of age)



INSECT REPELLENTS & INSECTICIDES



AGE

REPELLENTS	DEET	⚠ Not <6 months 6 months to 2 years: 10% DEET once daily 3 to 12 years: 10% DEET up to three times daily >12 years: Up to 30% DEET up to three times daily
	Icaridin	⚠ Not <6 months >6 months: Up to 20% icaridin
	PMD	⚠ Not <3 years >3 years: 10% PMD up to twice daily
	Citronella	⚠ Not <2 years >2 years: 5% to 15% citronella as needed
	Soybean Oil	Suitable for all ages 2% as needed
	Metofluthrin	Suitable for all ages ⚠ Not to be applied to skin. Keep out of reach of children.
INSECTICIDES	Permethrin	>16 years only ⚠ Not to be applied to skin. Keep out of reach of children.



NON-PHARMACOLOGICAL TREATMENT

- Clean area with mild soap and warm water
- For bee stings:
 - Remove the stinger to reduce the local reaction
 - Use a credit card or flat edge to scrape the stinger, as tweezers may squeeze additional venom into the skin
- Ice or cool compresses to the skin may reduce inflammation
- Toothpaste, vinegar, and salt have limited favourable evidence



Note: Patients should be advised to avoid scratching or manipulating insect bites due to risk of infection



PHARMACOLOGICAL TREATMENT OPTIONS

MEDICATION	EXAMPLES	SIDE EFFECTS AND CONSIDERATIONS
First-generation antihistamines	• Chlorpheniramine • Cyproheptadine • Diphenhydramine • Hydroxyzine	• S/E: sedation, cognitive impairment, anticholinergic effects • Shorter duration of action
Second-generation antihistamines	• Bilastine • Cetirizine • Desloratadine • Fexofenadine • Loratadine • Rupatadine	• Lower risk of side effects compared to first-generation antihistamines • Longer duration of action (once-daily dosing)
Oral analgesics	Acetaminophen	Not normally needed for treatment of urticaria alone



PHARMACOLOGICAL TREATMENT OPTIONS

MEDICATION	ADMINISTRATION	SIDE EFFECTS AND CONSIDERATIONS
Lidocaine (topical anesthetic)	0.5% to 9.6% cream, ointment, lotion, gel, or spray - Apply up to 3 times daily as needed	- S/E: skin irritation and sensitization - Risk of methemoglobinemia in children - Maximum use of 7 consecutive days
Hydrocortisone (topical corticosteroid)	0.5% to 2.5% cream, ointment, or lotion - Apply as directed by the product information; frequency of administration varies according to the strength of the topical product	- S/E: skin irritation, dryness, or cracking - Maximum use of 7 consecutive days - Avoid use near mucous membranes
Menthol (topical counterirritant)	0.5% to 1% in various topical preparations or specially compounded - Apply sparingly as needed	- S/E: skin irritation, burning, redness - Avoid use near mucous membranes and broken skin
Calamine (topical astringent/skin protectant)	1% to 16% in lotion with zinc oxide - Apply liberally as needed	- S/E: may leave a chalky residue that is irritating and drying to the skin - Keep liquid away from children to avoid accidental ingestion



SPECIAL POPULATIONS

PREGNANCY AND LACTATION

- Non-chemical protection, such as physical barriers and avoiding infested areas preferred
- Evidence of harm to unborn or breastfed babies from the use of DEET, icaridin, or PMD by the mother has been lacking, and therefore these insect repellents can be considered
- Chemical protection against mosquito bites is recommended when pregnant women enter areas with a significant risk of diseases linked to insect bites, such as malaria or Lyme disease
- Citronella is NOT recommended in pregnancy
- Soybean oil is considered to be safe for use during pregnancy and breastfeeding



MONITORING

- Pain
- Fever
- Inflammation
- Itch
- Systemic symptoms
- Delayed anaphylaxis
- Infection

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

AT is pregnant in her first trimester and would like to know which insect repellent(s) she can use while taking walks around her local park. Which of the following are acceptable options for AT? Select all that apply.

- ☐ a) 30% DEET applied up to three times daily
- ☐ b) 20% icaridin applied once daily
- ☐ c) 10% oil of lemon eucalyptus (PMD) applied once daily
- ☐ d) 15% oil of citronella applied every 2 hours as needed

Quiz

Click the **Quiz** button to edit this object

CHECK YOUR UNDERSTANDING

Is the following statement true or false?

Itching related to an insect bite should be monitored for 7 days.

☐ True

☐ False



CLINICAL PEARLS

- Attempt to identify the type of insect as management may vary
- Signs of anaphylaxis require immediate medical treatment
- Recommend low strengths of insect repellent for children and pregnant women
- Avoid topical diphenhydramine due to its risk of causing contact dermatitis or sensitization
- Apply insect repellent onto exposed skin and clothes, 20 minutes after sunscreen
- Wash hands after applying insect repellent and wash insect repellent off when protection is no longer needed
- Ensure patients with a history of anaphylaxis to insect bites are counselled on the use of epinephrine

PATIENT CASE



CASE BACKGROUND

Background: JY is a 7-month-old male who is brought into the clinic by his mother. She states they are going camping this upcoming weekend and would like to know more about the use of insect repellents for prevention of insect bites as well as treatment options for infants.



PAST MEDICAL HISTORY

- Caesarean delivery
- Otherwise unremarkable

ALLERGIES

- Latex (rash)

SOCIAL HISTORY

- Breastfed and slowly introducing solid foods

MEDICATIONS

- Vitamin D drops 400 units daily



CASE QUESTIONS

1. Which ONE of the following statements is true?
 - a) DEET in strengths of up to 20% may be used in children over 2 months of age
 - b) DEET in strengths of up to 10% may be used in children over 2 months of age
 - c) The use of icaridin is safe for children older than 6 months of age
 - d) Use of citronella in children under 2 years of age is safe and effective

2. Which ONE of the following would be an appropriate non-pharmacological treatment option for JY if they were to get an insect bite?
 - a) Apply petrolatum on the affected area to reduce swelling
 - b) Pop any blisters that arise due to the insect bite
 - c) Apply lidocaine to the affected area for local anesthesia
 - d) Use warm water and mild soap for symptom management



CASE QUESTIONS

3. Which ONE of the following natural products can be recommended as insect repellent for JY?
- a) Citronella Oil
 - b) Soybean Oil
 - c) Lavender Oil
 - d) Sage Oil
4. Which topical product is **NOT** recommended for use for an insect bite on JY's arm?
- a) Hydrocortisone
 - b) Diphenhydramine
 - c) Warm water and mild soap
 - d) Calamine



CASE SOLUTIONS

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REFERENCES



Or click [here](#)
to see the
reference list